

ADVANCED MICROECONOMETRICS
Problem Set - Day 2

SERGIO URZUA
DUE DATE: JULY 28, 2026

Instructor: Sergio Urzua

1. Consider the following model of potential utility levels:

$$U_1 = \mu_1 + \epsilon_1 \quad (\text{withdrawing from 101 after the midterm exam})$$

$$U_0 = \mu_0 + \epsilon_0 \quad (\text{not withdrawing from econ 101 after the midterm exam})$$

where μ_i is an intercept and ϵ_i is the associated error term ($i = 0, 1$). Additionally, assume the following decision model:

$$D = \mathbf{1} [T < \bar{T}]$$

where $D = 1$ means “withdrawing after the midterm exam” $D = 0$ otherwise, T represents the midterm score, and $\bar{T}$ is a threshold. Assume the following production function for the score:

$$T = \gamma Z + V_D$$

where Z is a vector of controls (including hours of study before the midterm), and V_D is an error term. Assume Z and V_D independent, Z and ϵ_1 independent, and Z and ϵ_0 independent. Finally, assume that $(V_D, \epsilon_1, \epsilon_0)$ are jointly normally distributed with mean $(0, 0, 0)$ and variance-covariance matrix Σ .

- (a) Explain why the sign of $E(U_0 - U_1 | D = 0, Z)$ determines if, on average, people solving this exam made no mistake by staying in the class after the midterm.
- (b) Show that, on average, you made the right decision (i.e., $E(U_0 - U_1 | D = 0, Z) > 0$) as long as:

$$\frac{(\mu_0 - \mu_1)}{\lambda \left(\frac{\bar{T} - \gamma Z}{\sqrt{\text{Var}(V_D)}} \right)} > \left(\frac{\text{Cov}(\epsilon_1, V_D) - \text{Cov}(\epsilon_0, V_D)}{\sqrt{\text{Var}(V_D)}} \right)$$

where $\lambda \left(\frac{\bar{T} - \gamma Z}{\sqrt{\text{Var}(V_D)}} \right) = \frac{\phi \left(\frac{\bar{T} - \gamma Z}{\sqrt{\text{Var}(V_D)}} \right)}{1 - \Phi \left(\frac{\bar{T} - \gamma Z}{\sqrt{\text{Var}(V_D)}} \right)}$, and $\phi(\cdot)$ and $\Phi(\cdot)$ represent the pdf and cdf of a standard normal random variable, respectively.

- (c) For simplicity assume $\mu_0 = \mu_1$, so you made the right decision as long as $Cov(\epsilon_1, V_D) < Cov(\epsilon_0, V_D)$. How plausible is this case? What should be the signs of $Cov(\epsilon_1, V_D)$ and $Cov(\epsilon_0, V_D)$? Interpret your results.
- (d) Suppose you have access to information on utility levels for the sample of individuals that originally enrolled in econ 101. Suppose that you also know whether or not each of the individuals in the sample dropped the class (you observe D). Show that you can use the previous set-up (assuming $\mu_0 \neq \mu_1$) to justify the following equation
- $$U_i = \alpha + \beta D_i + \epsilon_i \quad (1)$$
- where U_i is observed utility.
- (e) Should you run OLS in (1) to estimate the effect of withdrawing from econ 101 on utility levels?
- (f) Explain why “hours of study before the midterm exam” would be a valid instrument in this set-up.
- (g) Assume $\epsilon_1 = \epsilon_0$. Explain how you would use “hours of study before the midterm” to get the average effect of withdrawing from econ 101.
- (h) Assume $\epsilon_1 \neq \epsilon_0$. Explain how you would use “hours of study before the midterm” to get the average effect of withdrawing from econ 101.

2. Consider the model:

$$D = 1[\gamma_0 + \gamma_1 Z > V].$$

and assume you observe $Y_{i,1} = \alpha_1 + \beta_1 X_i + \epsilon_{i,1}$ if $D = 1$, you observe $Y_{i,0} = \alpha_0 + \beta_0 X_i + \epsilon_{i,0}$ otherwise. Assume the following parameterization: $Z \sim N(0, 2)$, $X \sim N(0, 1)$, $\epsilon_1 \sim N(0, 1)$, $\epsilon_0 \sim N(0, 2)$. $V \sim N(0, 1)$, $\alpha_0 = 0.15$, $\alpha_1 = 0.25$, $\beta_0 = -0.55$, $\beta_1 = 0.45$, $\gamma_0 = 0.25$, and $\gamma_1 = -1$. V , ϵ_0 and ϵ_1 are mutually independent.

- (a) Generate a fake sample of 1,000 observations.
- (b) Code the associated likelihood function and estimate the parameters of the model using information on (Y, D, Z, X) . Compare your results with the true values.
- (c) Compute standard errors for your estimates.
- (d) Repeat the previous analysis but assuming $V \sim N(0, 2)$. What are the implications?
- (e) Repeat the previous analysis but assuming $\epsilon_1 = 0.2 \times V$, $\epsilon_0 = -0.2 \times V$ where $V \sim N(0, 1)$.

- i. Describe the sorting patterns.
- ii. Presents your estimated OLS, IV, ATE and TT.
- iii. Explain your biases.